

HOLY FAITH ACADEMY, DHULIYAN
Affiliated to CISCE, New Delhi
Semester-I Examination
Session: 2025-26
Class IX
Biology

TSC/E

Total Marks: 80 Time: 2 hrs

General Instructions:

1. Answers to this paper must be written on the paper provided separately.
2. You will not be allowed to write during the first 15 minutes. Use this time in reading the question paper.
3. Attempt all questions from Section I and any four questions from Section II. The intended marks of questions or parts of questions are given in brackets [].

SECTION-I (40 Marks)

[Attempt all questions from this section.]

Question [a to g - all compulsory]

(a) Name the following: [5]

- (i) Air filled parenchyma needed for buoyancy in aquatic plant.
- (ii) Erythropoietin secreting organs detox our urea from blood.
- (iii) Discoidal cells, i.e., rich in oxygen binding protein.

(iv) A structure in stamen connects two lobes of an anther.

(v) Fat emulsification is done by this hepatic secretion.

(b) Select one suitable term from the brackets and fill in the blanks: [5]

(i) The female reproductive part of flower is called _____. [Carpel, stamen, corolla]

(ii) Vallisneria is pollinated by _____. [water, air, animals]

(iii) _____ is an exception to modern cell theory. [HIV, Bacteria, Eukaryotes]

(iv) Cardiac muscle is _____. [uni/multinucleated]

(v) Chromosomes are made up of protein and _____. [Histone, DNA, RNA]

(c) State one point of difference between the following pairs on the basis of what is indicated in the brackets. [5]

(i) Parasitic and Saprophytic nutrition [definition]

(ii) RER and SER [Function/location]

(iii) Blood & Lymph [any difference]

(iv) Epigeal vs Hypogeal germination [define/growth of epi/hypocotyl]

(v) Tendon & ligament [location]

(d) Define the following: [5]

(i) Pollination

(ii) Vivipary

(iii) Complete flower

(iv) Animalcules

(v) Peristalsis

(e) State whether the following are True or False. Correct and rewrite the statements if False. [5]

(i) Cross-pollination requires abiotic or biotic agents.

- (ii) Stone cell or grit cell is found in hard shelled animal.
- (iii) Plumule is part of embryo.
- (iv) Fats are non-essential components of food.
- (v) All prokaryotic cells are useful to man.

(f) Match A & B column: (just write the matches answer) [5]

A	B
<i>(To be completed by student)</i>	

(g) Give one example/answer of each: [5]

- (i) Two exceptions to cell theory.
- (ii) Matrix with protein complex chondrin.
- (iii) Iron [Fe⁺⁺] containing protein in RBC.
- (iv) Black reaction for cellular visualization developed by C. Golgi.
- (v) ATP synthesizing machinery called oxysome present in organelle.

SECTION-II (40 Marks)

[Attempt any four.]

Question 1 [10]

- (a) Design an experiment performed on germinating seeds to display oxygen, temperature, and light are needed. [3]
- (b) How do you prove that germinating seed takes oxygen, not CO₂ gas? Should you conduct Hg-capillary tube experiment? [3]
- (c) Write the chemical equation for the process. [2]
- (d) Write the three functions of Golgi apparatus. [2]

Question 2 [10]

- (a) Explain the different types of placentation with examples. [4]

(b) Name the first enzyme and one natural antibiotic from saliva mixed with food in holozoic nutrition. [3]

(c) Why does an autotroph not involve the digestion process? [3]

Question 3 [10]

(a) Explain agents of pollination. [5]

(b) Contrast flowers displaying Entomophily and Anemophily. [5]

Question 4 [10]

(a) Draw a neat structure of prokaryotic cell. [4]

(b) Mention any two organelles absent in it. [3]

(c) Are plastid, lysosome, and ribosome present in all cells? If not, mention the cell that lacks these structures. [3]

Question 5 [10]

(a) Design a single experiment to prove that air, water, and temperature are needed for the germination of seeds. [6]

(b) Why do you think oxygen and water are required for seed germination? [4]

Question 6 [10]

(a) Draw the structure of a maize grain. [4]

(b) Explain the mode of germination in Castor with a simple diagram. [3]

(c) What is caruncle? [3]